

Supplementary methods

A configurable generative transcriptomics framework for spaceflown mice

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Supplementary methods. This document records the data roles, model architecture, evaluation gates, statistical safeguards, and complete supporting results.

S1. OSDR discovery and expression ingestion

OSDR data were obtained through the Biological Data API documented at <https://visualization.osdr.nasa.gov/biodata/api/>.

Eligibility required:

- `Mus musculus`;
- transcription profiling by bulk RNA sequencing;
- a resolvable spaceflight/flight or ground-control label;
- processed RSEM expected-count output;
- sample-level tissue or material metadata sufficient for canonicalization.

No raw combined OSDR HDF5 file was used. The API returned 1,631 profile rows. Twenty-one technical replicates were aggregated, leaving 1,610 biological profiles, 835 flight and 775 ground control, from 75 accessions and 24 canonical material classes.

S2. ARCHS4 cohort audit

The source file was `assets/archs4/mouse_gene_v2.5.h5`, containing 997,515 profiles and 53,511 genes.

The three eligible reference cohorts contained:

Cohort	Profiles	GEO series	Intended use
Control only	23,614	3,213	Conservative sensitivity cohort
Healthy preferred	62,299	5,307	Primary pretraining source
Broad	134,250	15,111	Diversity sensitivity cohort

Before reference selection, nine GEO series linked to eligible OSDR accessions were excluded by accession metadata. This removed 108 otherwise eligible ARCHS4 profiles, including all 53 selected profiles from GSE152382 (OSD-457). The paper-parity run then selected 17,244 healthy-preferred profiles across 20 tissue classes. Complete GEO-series grouping assigned 10,150 profiles to training, 2,466 to validation, and 4,628 to test, with no series shared across splits.

S3. Landmark panel and normalization

Full-transcriptome TPM was calculated with GENCODE vM39 mouse gene lengths. Landmark selection occurred after TPM calculation and used a deterministic 974-gene mouse mapping of the human L1000 panel. Training-set MaxAbs scaling was applied after landmark selection.

S4. ARCHS4 DDIM configuration

The full configuration is `configs/rna_diffusion/archs4_mouse_paper_parity_osdr_disjoint.yaml`. The neural architecture and optimizer settings match the paper-parity implementation; only the reference exclusion and grouped split contract changed.

Component	Value
Input genes	974
Hidden layers	8,192; 8,192
Parameters	227,109,786
Dropout	0.1
Diffusion steps	1,000
Beta schedule	Quadratic, 0.0001 to 0.02
Objective	Summed noise MSE
Optimizer	Adam
Learning rate	0.0004783833151836702
Scheduler	OneCycle
Batch size	2,048
Epochs / optimizer steps	15,000 / 75,000
AMP	Enabled
EMA	0.999
Device	NVIDIA A100-SXM4-40GB
Runtime	6,083 seconds
Peak allocated GPU memory	5.93 GB

S5. Factorized OSDR adaptation

The accepted configuration is `configs/rna_diffusion/osdr_factorized_study_lora512_correlation_refine_osdr_disjoint.yaml`. It uses the ARCHS4 backbone described in Section S4. A base adapter was trained for 12,000 domain and 4,000 condition steps, followed by the 4,000-domain-step and 1,000-condition-step correlation-refinement stage reported here. The all-tissue and muscle-group development screens were then regenerated from this adapter.

Component	Value
Backbone	Completed ARCHS4 paper-parity DDIM
Conditioning	Tissue, FLT/GC, accession, material type
Domain adapter	LoRA rank 512, alpha 512
Domain stage	4,000 steps, LR 0.00002
Condition stage	1,000 steps, LR 0.00002
Batch size	512
Condition dropout	0.15
Correlation regularization	Weight 10, 256 genes, timestep ≤ 200
Sampling	100 DDIM steps for evaluation

The data split contained 781 training, 536 validation, and 293 locked-test profiles. Every test accession was represented in training. Results therefore measure within-study interpolation.

The accepted calibrator used:

1. train-only global alignment;
2. hierarchically shrunk accession and tissue means;
3. positive missing-covariance residual noise;
4. no condition-specific fit of calibrator means or covariance;
5. explicit clipping at zero for accepted downstream expression.

S6. Distribution and condition gates

Four generation seeds, 5020-5023, were declared before the locked test was opened. Every metric was gated independently.

Metric	Gate
Gene-correlation agreement	Paper target ≥ 0.98 ; finite-sample real-bootstrap floor also reported
Precision	≥ 0.95
Recall	≥ 0.85
F1	≥ 0.90
Adversarial accuracy	0.40 to 0.60
FD / real-split P95	≤ 1.0
Pooled effect recovery	Correlation ≥ 0.30 and direction ≥ 0.55
Muscle accession recovery	Correlation ≥ 0.30 and direction ≥ 0.55
Memorization	Generated fraction below train LOO P01 ≤ 0.05

The exact repeat rows are in `source_data/table_s2_locked_ddim_repeats.tsv`. On the locked within-study test, mean gene-correlation agreement was 0.9744, precision 0.9974, recall 0.9957, F1 0.9966, adversarial accuracy 0.4753, and FD/real-split-P95 ratio 0.0740. All four repeats passed the finite-sample correlation floor of 0.9497, although the mean remained below the paper target of 0.98. Pooled FLT/GC effect recovery passed in three of four repeats and muscle accession-effect recovery passed in all four. The preceding validation screen missed its stricter sample-specific correlation floor and muscle accession-effect gate; the broad biological screens therefore remain developmental.

The term "adversarial accuracy" refers to an external nearest-neighbor real-versus-synthetic classifier, not the WGAN training critic. A result near 0.5 indicates that this external discriminator cannot reliably separate the two cohorts.

S7. Configurable benchmark and comparator models

The resolved planner contains 463 gated experiment rows. This is a staged plan, not evidence that all 463 rows received paper-duration training. Smoke tests, paper-architecture feasibility runs, preprocessing screens, accession-scope tests, harmonization comparisons, study-conditioning experiments, and tissue expansion were advanced only when the preceding gates supported the next cost. The framework separates input units, library normalization, transformation, scaling, feature space, harmonization, training regime, accession scope, tissue mode, FLT/GC conditioning, study policy, balancing, seed, and synthetic-to-real ratio.

Liver harmonization benchmark

Nine arms used the same 974-gene conditional DDIM architecture, seed, and split: 119 training profiles, 50 validation profiles, and a 70-profile locked test that was not opened for this screen. Every arm ran on an NVIDIA A100. No arm passed all independent fidelity, pooled-condition, and accession-effect gates.

Method	Corr.	Precision	Recall	F1	AA	FD/real P95	Fidelity gate	Accession gate
No harmonization, TPM	0.283	0.200	0.960	0.331	0.850	2.052	Fail	Fail
lIangovan study z-score	0.278	0.160	1.000	0.276	0.770	0.716	Fail	Fail
Mentor two-stage z-score	0.348	0.440	1.000	0.611	0.690	0.977	Fail	Fail
ComBat	0.004	0.040	1.000	0.077	0.810	63.185	Fail	Fail
ComBat-seq	0.067	0.020	1.000	0.039	0.850	156.647	Fail	Fail
MBatch Median Polish	0.009	0.020	1.000	0.039	0.930	205.470	Fail	Fail
MBatch Empirical Bayes	0.001	0.000	1.000	0.000	0.850	60.625	Fail	Fail
MBatch ANOVA	-0.003	0.020	1.000	0.039	0.870	44.716	Fail	Fail
MOBER	0.808	0.260	1.000	0.413	0.770	33.311	Fail	Fail

ComBat, ComBat-seq, and the three MBatch validation transformations used training anchors but remain transductive sensitivity analyses for a new batch. MOBER was the inductive complex harmonizer. Its high correlation did not compensate for low precision and F1, external separability, or excessive distributional distance. Exact values and preprocessing labels are in Tables S13-S14.

WGAN-GP and GeneJEPA screens

The WGAN used the Viñas et al. topology: 64-dimensional noise, two 256-unit generator and critic layers, five critic updates, gradient-penalty weight 10, RMSProp learning rate 0.0005, and batch size 32. The strongest study-conditioned validation result was evaluated across six sampling seeds. Mean correlation, precision, recall, and F1 were 0.9759, 0.9764, 0.9938, and 0.9850. Mean external adversarial accuracy was 0.6362 and FD/real-P95 ratio was 0.1439. Pooled FLT/GC effect recovery passed in six of six repeats, but accession-aware skeletal-muscle recovery passed in zero of six. Because no repeat passed the full fidelity gate and accession-aware recovery was unstable, its locked test remained unopened. Exact repeat rows are in Table S15.

The exact-architecture GeneJEPA duration screen used 4,096 genes and 43,744 replacement-sampled training exposures. It reached 0.703 held-out tissue balanced accuracy versus 0.839 from expression. It is representation-only and has no expression decoder.

S8. Generated-feature workflow

The evaluation funnel had three stages: pooled utility, tissue-specific development, and real-data association testing. The pooled utility benchmark compared real-only, generated-only, and real-plus-generated training. Tissue-specific development then compared five arms:

1. `real_only`;
2. `generated_only`;
3. `real_plus_generated`, equal total real and synthetic weight;
4. `guided_real_only`, real classifier with real/synthetic consensus ranking;
5. `guided_low_weight`, real plus recentered synthetic profiles at 0.05 total synthetic weight.

Splits were nested within accession-by-condition strata. The inner loop selected feature count, regularization, and rank method. The outer loop measured balanced accuracy, AUROC, and average precision separately. No composite score was used. Because the same accessions could contribute

profiles to training and testing, this stage measured within-study development utility. It did not test transfer to a new study.

Stable genes had selection frequency at least 0.50 and coefficient-sign agreement at least 0.75. Generated-supported status did not constitute biological evidence. Real random-effects and LOO tests were applied afterward.

For the BH-FDR synthetic-informed subset, `reinforced` denotes a `core_intersection` gene that was stable in both the real-only and selected synthetic-guided arms, had matching coefficient directions, and had a supporting real meta-effect. `Synthetic-promoted` denotes a `generated_supported` gene that was stable in the selected synthetic-guided arm but did not cross the real-only stability threshold, with a supporting real meta-effect. Thus `synthetic-promoted` is a feature-selection classification, not a claim that the gene was absent from real expression, statistically nonsignificant in real data, or biologically novel.

All five arms were fitted for every analysis unit below. Values are means across eight repeated outer splits, and every outer evaluation used real profiles. An eligible arm was nonworse than real-only training in balanced accuracy, AUROC, and average precision. An eligible tie met that rule without improving a mean metric. These development results use profiles from represented accessions.

Supplementary Table S9. Complete canonical-tissue utility screen.

Tissue	n (FLT/ GC)	Selected arm	BA real/ selected	AUROC real/ selected	AP real/ selected	Status
Adrenal gland	31 (16/15)	Generated only	0.781 / 0.922	0.906 / 0.984	0.918 / 0.988	Eligible improvement
Bone	30 (15/15)	Guided ranking; real fit	0.656 / 0.734	0.797 / 0.828	0.858 / 0.865	Eligible improvement
Bone marrow	20 (10/10)	Generated only	0.969 / 1.000	1.000 / 1.000	1.000 / 1.000	Eligible improvement
Brain	45 (22/23)	Generated only	0.688 / 0.740	0.712 / 0.802	0.731 / 0.788	Eligible improvement
Brown adipose tissue	20 (10/10)	Generated only	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000	Eligible tie
Cecum	16 (8/8)	Real only	0.875 / 0.875	0.938 / 0.938	0.958 / 0.958	Real-only retained
Cerebellum	44 (23/21)	Generated only	0.529 / 0.602	0.621 / 0.696	0.729 / 0.764	Eligible improvement
Colon	45 (23/22)	Real only	0.656 / 0.656	0.677 / 0.677	0.730 / 0.730	Real-only retained
Eye	18 (9/9)	Generated only	0.594 / 0.781	0.562 / 0.781	0.729 / 0.865	Eligible improvement
Heart	42 (21/21)	Generated only	0.594 / 0.719	0.625 / 0.750	0.642 / 0.778	Eligible improvement
Hippocampus	30 (16/14)	Generated only	0.698 / 0.703	0.740 / 0.781	0.849 / 0.864	Eligible improvement
Kidney	111 (56/55)	Guided ranking; 5% synthetic	0.654 / 0.707	0.680 / 0.771	0.683 / 0.799	Eligible improvement
Liver	197 (101/96)	Real only	0.721 / 0.721	0.764 / 0.764	0.773 / 0.773	Real-only retained
Lung	63 (32/31)	Generated only	0.664 / 0.742	0.680 / 0.830	0.684 / 0.833	Eligible improvement
Mammary gland	20 (8/12)	Generated only	0.771 / 0.885	0.854 / 0.958	0.875 / 0.958	Eligible improvement
Optic nerve	29 (19/10)	Guided ranking; 5% synthetic	0.769 / 0.819	0.863 / 0.950	0.950 / 0.983	Eligible improvement
Retina	63 (37/26)	Guided ranking; 5% synthetic	0.587 / 0.723	0.620 / 0.762	0.756 / 0.846	Eligible improvement
Skeletal muscle, pooled	149 (74/75)	Guided ranking; real fit	0.861 / 0.932	0.924 / 0.961	0.908 / 0.945	Eligible improvement
Skin	125 (66/59)	Real + generated	0.671 / 0.756	0.691 / 0.768	0.747 / 0.808	Eligible improvement
Spleen	86 (44/42)	Real + generated	0.517 / 0.648	0.540 / 0.704	0.589 / 0.749	Eligible improvement
Thymus	94 (51/43)	Guided ranking; 5% synthetic	0.733 / 0.844	0.818 / 0.887	0.862 / 0.918	Eligible improvement
White adipose tissue	20 (10/10)	Generated only	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000	Eligible tie

Supplementary Table S6. Complete anatomical muscle-group utility screen.

Tissue	n (FLT/GC)	Selected arm	BA real/ selected	AUROC real/ selected	AP real/ selected	Status
EDL	24 (12/12)	Real only	0.917 / 0.917	1.000 / 1.000	1.000 / 1.000	Real-only retained
Gastrocnemius	25 (10/15)	Guided ranking; 5% synthetic	0.604 / 0.646	0.646 / 0.750	0.731 / 0.798	Eligible improvement
Quadriceps	35 (18/17)	Real only	0.750 / 0.750	0.880 / 0.880	0.916 / 0.916	Real-only retained
Soleus	41 (22/19)	Real + generated	0.925 / 0.963	0.980 / 0.980	0.980 / 0.986	Eligible improvement
Tibialis anterior	24 (12/12)	Real + generated	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000	Eligible tie

Sample counts are shown as total development profiles followed by flight/ground-control counts. Small cohorts and ceiling-level scores remain exploratory even when a synthetic arm is eligible.

S9. Random-effects reporting and LOO sensitivity

For gene (g) in accession (a), the real flight effect was:

```
delta[g,a] = mean(real expression[g] | FLT,a)
             - mean(real expression[g] | GC,a)
```

Accession effects were combined with a random-effects model. Within each tissue, Benjamini-Hochberg (BH) adjustment was applied to the 974 gene-level meta-analysis P values. BH sorts the P values and controls the expected proportion of false discoveries among the rejected hypotheses. The primary inclusion rule was BH FDR < 0.05; synthetic selection, study-direction agreement, heterogeneity, and LOO stability were not additional significance gates.

After BH correction, each association was annotated as:

- reinforced by stable real-only and synthetic-guided selection;
- synthetic-promoted;
- selected by the real-only arm;
- synthetic-selected without real-direction support; or
- not stably selected by either feature-selection arm.

The accession-direction fraction, random-effects variance (τ^2), heterogeneity (I^2), and exact study counts remain in the complete inventory. A gene with mixed study directions remains BH-significant when its pooled random-effects test passes, but the disagreement qualifies interpretation of the pooled effect.

The LOO analysis removed each accession, repeated the random-effects fit, and retained the maximum FDR and any sign reversal. LOO was treated as a sensitivity label rather than an inclusion requirement. A LOO-stable gene additionally required maximum LOO FDR < 0.05 and no LOO sign reversal. Failure of this label does not remove a gene from the primary BH-FDR table.

Generated profiles were never included in the random-effects model.

S10. Reactome analysis

The official mouse GMT was generated from `ReactomePathways.txt` and `Ensembl2Reactome_All_Levels.txt`, restricted to *Mus musculus*, R-MMU-* pathways, and ENSMUSG* genes.

Hypergeometric enrichment used the 974-gene landmark panel as background. Benjamini-Hochberg FDR was applied separately by tissue and selected-gene set. Reactome parent and child terms overlap. Counts of significant rows are therefore not counts of independent biological discoveries.

S11. Skeletal-muscle group analysis

The factorized DDIM and three frozen synthetic development views were reused. No additional neural network was trained for the muscle-group screen.

Group	Profiles	Accessions	FLT	GC
EDL	24	2	12	12
Gastrocnemius	25	3	10	15
Quadriceps	35	4	18	17
Soleus	41	3	22	19
Tibialis anterior	24	2	12	12

The selected soleus arm was real plus generated expression. It improved mean balanced accuracy from 0.9250 to 0.9625, left AUROC at 0.9800, and increased average precision from 0.9804 to 0.9865. Five BH-FDR genes were reinforced by both stable real-only and selected-arm feature ranking and had the same real effect direction in all three accessions: FLT-lower *Bdh1*, *Ech1*, *Bnip3*, and *Decr1*, and FLT-higher *Tpm1*. *Bdh1*, *Ech1*, *Bnip3*, and *Tpm1* also passed the LOO sensitivity criterion; *Decr1* did not. The screen did not promote *Mef2c* or *Pxmp2*.

Gastrocnemius selected a guided low-weight arm and promoted *Fhl2* and *Nfkb1a*. Tibialis anterior selected real plus generated expression and reinforced *Cdkn1a*, *St3gal5*, and *Bnip3* while promoting *Cebpd*. None of these gastrocnemius or tibialis genes passed the LOO FDR sensitivity rule. EDL and quadriceps retained real-only arms, so their BH-FDR genes are not attributed to synthetic guidance.

LOO here is a real-data meta-analysis sensitivity test. It does not remove the accession from the already completed generator adaptation. Soleus remains developmental until a new accession is excluded from adaptation and selection.

S12. Random-effects BH-FDR gene inventories

Supplementary Table S11 is the primary statistical inventory. It contains every real-data random-effects association with BH FDR < 0.05, without requiring synthetic selection, unanimous study direction, or LOO stability. Supplementary Table S12 reports the corresponding counts for every tested analysis unit, including tissues with zero BH-FDR genes.

The 459 retained tissue-gene results comprise 202 associations across 10 of 22 canonical tissues and 257 across all five anatomical muscle groups. This is an inventory size, not a count of independent discoveries: genes can recur across tissues, and pooled skeletal muscle overlaps the anatomical

subgroup analyses. Of these results, 363 had the same effect direction in every represented accession and 96 did not. Direction agreement is reported as an annotation, not an exclusion rule.

Analysis unit	BH-FDR genes	FLT higher	FLT lower	Unanimous direction	Synthetic-promoted	Reinforced
Adrenal gland	22	1	21	21	1	1
Eye	8	4	4	8	0	1
Heart	4	2	2	4	0	0
Kidney	3	3	0	1	1	1
Liver	19	8	11	0	0	0
Retina	5	5	0	5	0	0
Skeletal muscle, pooled	26	12	14	5	4	8
Skin	3	2	1	1	1	0
Spleen	34	32	2	21	3	1
Thymus	78	37	41	57	13	3
EDL	136	14	122	130	0	0
Gastrocnemius	15	8	7	13	2	0
Quadriceps	29	25	4	22	0	0
Soleus	29	5	24	27	0	5
Tibialis anterior	48	42	6	48	1	3

Supplementary Table S10 is the 49-row synthetic-informed subset of the primary inventory: 26 synthetic-promoted and 23 reinforced tissue-gene results with supporting real effects. Synthetic attribution is suppressed when the candidate generated arm failed the three-metric eligibility gate. The table is retained to describe what the generator changed in feature ranking, but it is not the primary significance table. The full inventory additionally contains 34 real-only selected results, 370 BH-significant results not stably selected by either arm, and six synthetic-selected results without real-direction support.

The complete synthetic-informed gene set is shown below. FLT higher and FLT lower refer to the sign of the real random-effects meta-estimate. An asterisk marks a BH-significant pooled effect whose direction was not identical in every accession.

Synthetic-promoted genes

Analysis unit	FLT higher	FLT lower
Adrenal gland	None	Psmb8
Kidney	Inpp4b*	None
Skeletal muscle, pooled	None	Klhl21*, Mapkapk5*, Reep5*, Itgb5*
Skin	Plscr1	None
Spleen	Rai14, Myl9*, Ptpkr	None
Thymus	Hsd17b11*, Etv1	Nusap1, Stmn1, Birc5, Cdk1, Top2a, Ccnb2, Aurka, Ccne2, Kif20a, Pcna, Ccnf
Gastrocnemius	Nfkb1a	Fhl2
Tibialis anterior	Cebpd	None

Reinforced genes

Analysis unit	FLT higher	FLT lower
Adrenal gland	None	Tspan4
Eye	None	Klhl21
Kidney	Slc37a4	None
Skeletal muscle, pooled	Sox4, Cebpd*, Sh3bp5, Prkcd, Arid5b*, Sesn1*, Tle1*	Bphl*
Spleen	Loxl1	None
Thymus	Snx7	Ube2c, Gmn
Soleus	Tpm1	Bdh1, Ech1, Bnip3, Decr1
Tibialis anterior	Cdkn1a, St3gal5, Bnip3	None

Heart, liver, retina, EDL, and quadriceps had real-data BH-FDR genes but no synthetic-informed gene. Bone, bone marrow, brain, brown adipose tissue, cecum, cerebellum, colon, hippocampus, lung, mammary gland, optic nerve, and white adipose tissue had no BH-FDR gene in the 974-gene panel.

The pooled skeletal-muscle results remain auditable in Tables S11-S12 and are interpreted separately from anatomical groups because those groups have different responses. Liver has 19 real-data BH-FDR associations but selected a real-only arm. Skin has three real-data BH-FDR associations and one synthetic-promoted gene, *Plscr1*. Lung has no BH-FDR gene in the 974-gene panel.

S13. Supplementary figures

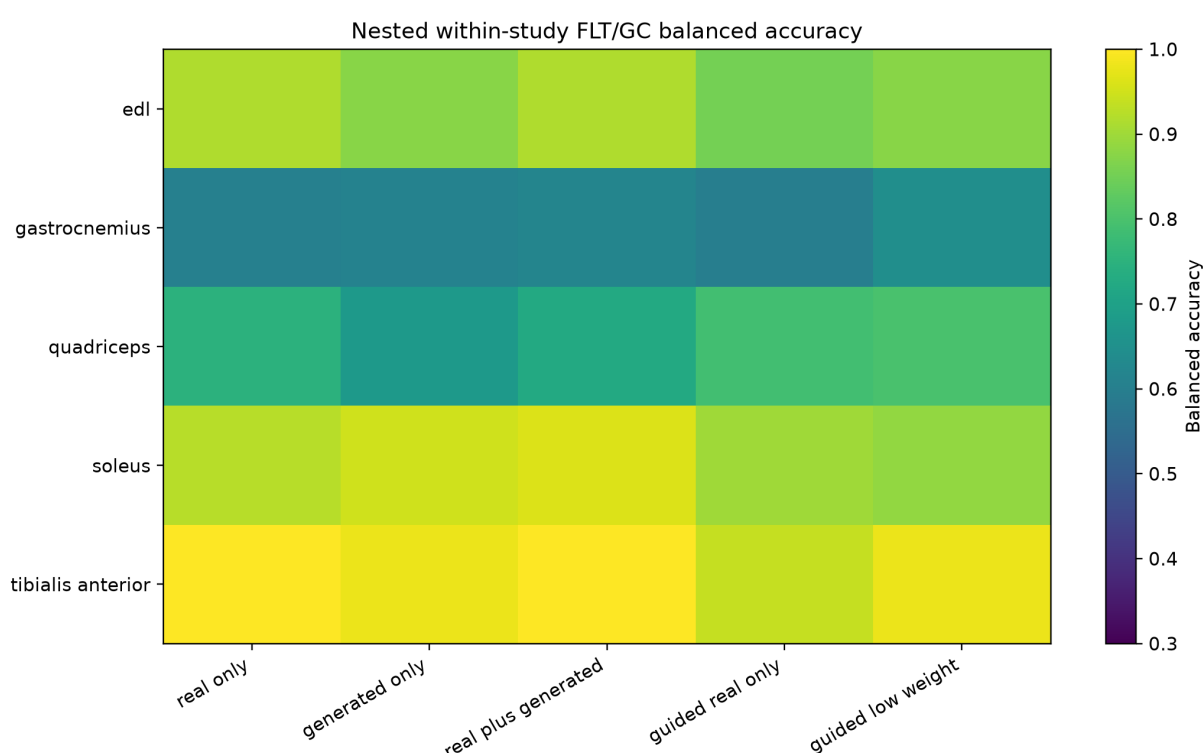


Figure S1. Repeated nested muscle-group balanced accuracy. Each row is a muscle group and each column is a downstream use of real or generated profiles. Arm selection also required nonworse AUROC and average precision.

Pooled augmentation failed, while tissue-specific synthetic use varied

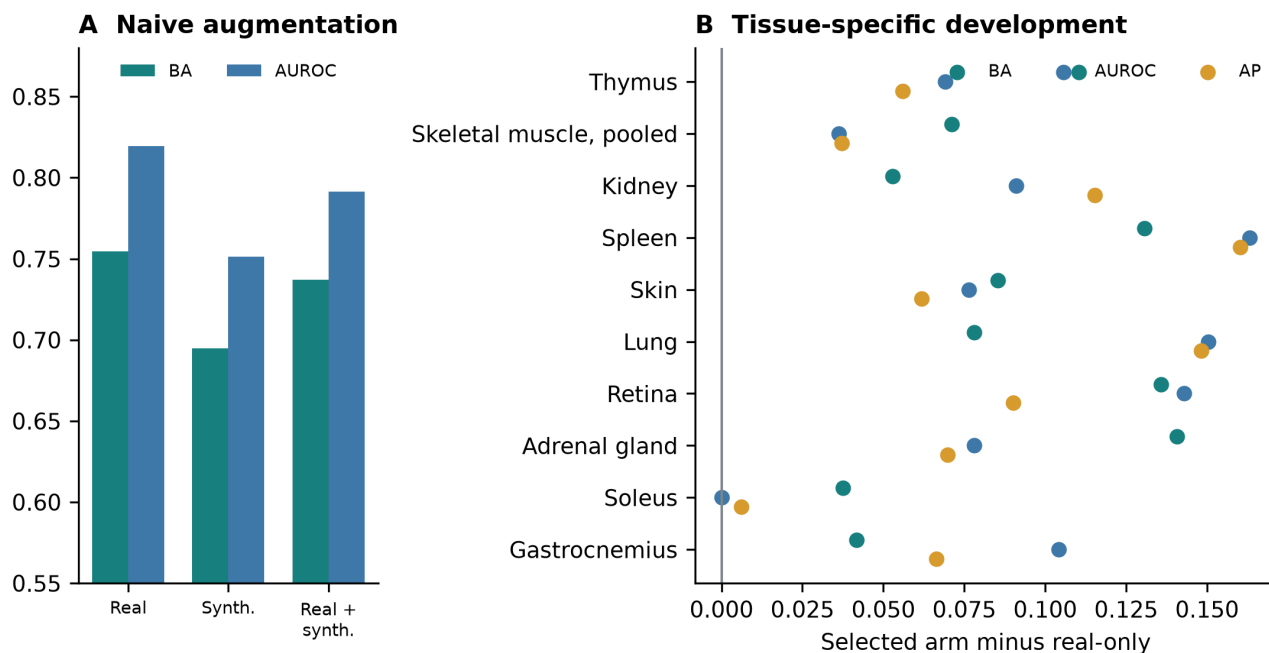


Figure S2. Downstream utility of generated expression. (A) Direct pooled augmentation on the locked real-profile test. (B) Selected-arm changes in balanced accuracy, AUROC, and average precision across repeated tissue-specific development splits. All evaluations used real profiles.

S14. Supplementary data tables

- `table_s1_archs4_ddim_metrics.tsv`
- `table_s2_locked_ddim_repeats.tsv`
- `table_s3_naive_augmentation.tsv`
- `table_s4_thymus_core_genes.tsv`
- `table_s5_thymus_reactome.tsv`
- `table_s6_muscle_group_summary.tsv`
- `table_s7_soleus_genes.tsv`
- `table_s8_muscle_reactome.tsv`
- `table_s9_all_tissue_development_screen.tsv`
- `table_s10_synthetic_informed_bh_fdr_genes.tsv`
- `table_s11_all_random_effects_bh_fdr_genes.tsv`
- `table_s12_bh_fdr_tissue_summary.tsv`
- `table_s13_liver_harmonization_benchmark.tsv`
- `table_s14_liver_harmonization_full_metrics.tsv`
- `table_s15_wgan_validation_repeats.tsv`